

Damodar Dasi

Senior Mechanical & Package Design Engineer with experience across compressor packages, PSA nitrogen systems, fuel systems, storage tanks, mechanical skids and integrated package design.



PROFILE
Mechanical & Package Design

EXPERIENCE
6+ years overall · 4+ compressor/package

LOCATION
Sharjah, UAE

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CORE SOFTWARE
SolidWorks · AutoCAD

PROFESSIONAL SUMMARY

I work at the intersection of mechanical design, package integration and project execution. My experience includes complete 3D package development, equipment arrangement, process and utility piping, lubrication systems, seal systems, structural/base frames, engineering drawings, MTO/BOM, multidisciplinary coordination, manufacturing support and FAT/SAT activities.

My project exposure includes international compressor packages, an ONGC PSA nitrogen generator, an Emarat mobile fueling unit, RFQ-stage fuel systems, storage tanks and mechanical/structural skids.

PROFESSIONAL EXPERIENCE

Nov 2025 — Present

Mechanical Design & Project Engineer

Meta Green Middle East FZ LLC · UAE

- Mechanical and package design for mobile and skid-mounted fueling systems, equipment packages, storage tanks and fabricated equipment.
- 3D models, GA/fabrication drawings and BOM/BOQ/MTO in SolidWorks and AutoCAD.
- Skid structures, equipment supports, piping, tanks, pumps and instrumentation; structural/mechanical calculations including weight, CoG and lifting considerations.
- Head of Design Department, Project Coordinator and Lead Mechanical Designer for the Emarat 40 FT dual-compartment MFU — tender through design, procurement, manufacturing, FAT and SAT.
- Same role on a demonstration portable fueling unit integrated with Maser Italia's smart fleet-management system, through manufacturing and commissioning.
- Approximately 25 storage tank designs from 500 L to 60 KL.

Apr 2023 — Oct 2025

Senior Design Engineer — Compressor, Compander & PSA Nitrogen Packages

DesignTech Systems · Atlas Copco onsite

- Led mechanical and package design for centrifugal compressor/compander packages and PSA nitrogen generator systems.
- Developed 3D assemblies in SolidWorks and 2D engineering/manufacturing drawings in AutoCAD.
- Worked on process piping, lubrication systems, seal piping, base frames, foundations, MTO/BOM and package integration.
- Led and trained a team of 6 designers/engineers: assigned design/drafting work, reviewed deliverables and drawings, and guided junior members.
- Supported customer/vendor coordination, manufacturing technical queries and FAT activities.

CORE EXPERTISE

- Compressor Packages
- PSA Nitrogen
- Fuel Systems
- Storage Tanks
- Mechanical Skids
- Package Integration
- Process Piping
- Lubrication Systems
- Seal Systems
- 3D Mechanical Design
- Manufacturing Drawings
- MTO / BOM
- FAT / SAT

SOFTWARE

- SolidWorks
- AutoCAD
- Autodesk Inventor
- AutoCAD 3D

LEADERSHIP & COORDINATION

- Led and trained a team of 6 designers/engineers on Atlas Copco package work.
- Head of Design Department, Project Coordinator and Lead Mechanical Designer for the Emarat 40 FT MFU at Meta Green UAE.
- Assigned modelling and drawing work and reviewed junior deliverables.
- Trained junior engineers/designers.
- Coordinated with senior management, project managers, procurement, production, instrumentation and customers/vendors.

RECOGNITION

- **Guiding Star** (mentorship) ×2 — DesignTech Systems, Q1 & Q3 2023, for mentoring 6 people.
- **Star Performer** ×2 — DesignTech Systems, Q2 & Q4 2022.

Mar 2022 – Mar 2023

Design Engineer — Compressor Packages

DesignTech Systems · Atlas Copco offsite

- Compressor package 3D design, GA layouts and manufacturing drawings.
- BOM/MTO, interfaces and package integration; equipment sizing and calculations.
- P&ID / equipment-selection coordination, vendor/customer coordination and FAT/SAT support.

Dec 2021 – Feb 2022

Design Engineer — Apprentice / Trainee

DesignTech Systems

Mechanical design support for industrial equipment, silencers, air-exhaust systems and fabricated structures using Autodesk Inventor.

Oct 2020 – Nov 2021

Junior Design Engineer

Sai Jyothi Engineering Works

Mechanical/piping design and assembly support, 3D modelling and detailed drawings with GD&T, and fabrication/inspection support.

SELECTED PROJECT EXPERIENCE

C-01 · NYK N₂ Comander Package

4-stage N₂ comander for shipyard service · Korea · 29,605 kg/h. Complete 3D package, lube-oil and seal-gas systems, base frame, process piping, water-cooled coolers (intercoolers, after cooler, LO and motor coolers), GA, MTO/BOQ, customer/vendor coordination, manufacturing support and FAT. API 617 · 614 · 650 · 692 · ASME B31.3 · ABS rules.

C-02 · Xiexing Rudong BOG Compressor

1-stage LNG boil-off gas compressor · LNG Rudong Co., China · 28,296 kg/h · 1.05 → 1.67 bar(a) · -158.3 °C inlet. Complete 3D SolidWorks model, AutoCAD layout/manufacturing/foundation drawings, lube-oil system with air cooler, seal piping and instrumentation integration; reviewed the analysis team's stress report; manufacturing/FAT support. API 617 · 614 · ASME B31.3.

C-03 · Banyan FGB Natural Gas Compressor

4-stage natural-gas compressor · Sembcorp Cogen, Singapore · 36,980 kg/h · 13.21 → 50.8 bar(a). Complete package redesign and integration: process piping calculations, lube-oil and seal-gas systems, air-cooled intercoolers and recycle cooler, noise hood, base skid, customer tie-ins and manufacturing/FAT support. API 617 · 614 · 520/521 · ASME B31.3.

C-04 · Sichuan H₂O₂ Compressor & Expander Package

2 × 3-stage air compressors + 2-stage expander on a common platform · China. Complete 3D arrangement, process piping, gas coolers, shared lube-oil system, lube-skid stress and foundation-load calculations, manufacturing/FAT support. API 672 · 617 · 614 · ASME B31.3.

C-05 · Maoming Air Compressor

3-stage steam-turbine-driven air compressor · China · 95,919 kg/h · 0.989 → 6.013 bar(a). Complete 3D package, filter-house inlet duct, process piping, water-cooled intercoolers, lube-oil system with overhead rundown tank; compressor-side coupling only (turbine customer-supplied); manufacturing/FAT support. API 672 · 614 · ASME B31.3.

P-01 · ONGC PSA Nitrogen Generator

160 Nm³/h N₂ · 99.9 % · ~6 barg · ISO 8573-1 [2:4:1]. Main mechanical/package designer: 3D design, sizing of compressor, aftercooler, separator, dryer, 12 m³ receiver and 25 m³ N₂ buffer (ASME VIII / IS 2825), package P&ID with the process engineer, MTO/BOM, FAT and some SAT.

F-01 · Emarat Mobile Fueling Unit

40 FT dual-compartment MFU-3A · 42,600 L gross tank volume · UAE. Head of Design Department + Project Coordinator + Lead Mechanical/Package Designer from tender through design, procurement, manufacturing, FAT and SAT.

APPLICABLE STANDARDS & CODES

Project experience includes work associated with API 614, API 617, API 672, API 692, ASME B31.3, ASME Section VIII, UL 142, UL 2085, NFPA, ATEX/IEC, ADR where applicable, and ISO 8573-1.

EDUCATION & LANGUAGES

B.Tech, Mechanical Engineering

Pragati Engineering College, JNTU Kakinada · 2016–2020

Industrial Training

45-day training, Vizag Steel Plant — Blast Furnace Operations

English · Hindi · Telugu